
DUETRAG: COLLABORATIVE RETRIEVAL-AUGMENTED GENERATION

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1 Abstract

Retrieval-Augmented Generation (RAG) methods augment the input of Large Language Models (LLMs) with relevant retrieved passages, reducing factual errors in knowledge-intensive tasks. However, contemporary RAG approaches suffer from irrelevant knowledge retrieval issues in complex domain questions (*e.g.*, HotPot QA) due to the lack of corresponding domain knowledge, leading to low-quality generations. To address this issue, we propose a novel Collaborative Retrieval-Augmented Generation framework, DuetRAG. Our bootstrapping philosophy is to simultaneously integrate the domain finetuning and RAG models to improve the knowledge retrieval quality, thereby enhancing generation quality. Finally, we demonstrate DuetRAG’s matches with expert human researchers on HotPot QA.

2 Introduction

Large language models (LLMs) have demonstrated significantly enhanced performance compared to preceding efforts within the realm of question answering (QA) (Achiam et al., 2023; Wei et al., 2022). Nonetheless, in the domain of domain-specific knowledge QA, LLMs frequently exhibit subpar performance attributable to an insufficiency of pertinent knowledge. LLMs, when fine-tuned on domain-specific knowledge, acquire specialized domain knowledge to address queries via fine-tuning on an extensive corpus of domain-specific information. However, this methodology typically necessitates copious amounts of data and entails substantial training overhead, while concurrently exposing the model to the risk of catastrophic forgetting (Kirkpatrick et al., 2017), thereby impeding its ability to achieve desired outcomes in QA across alternate domains. The paradigm of question answering grounded in retrieval-augmented generation (RAG) facilitates the integration of external knowledge into LLMs (Lewis et al., 2020), thereby enabling them to extract contextually pertinent information from external repositories in instances where the model lacks internal knowledge, thus improving the performance of large language models in domain-specific knowledge question answering. However, traditional RAG suffers from the illusion problem (Zhang et al., 2023), leading to incorrect answers, and when faced with more complex questions, limited by the performance of the retriever, RAG may encounter difficulties in answering based on poorly correlated or even erroneous external documents.

In this paper, we propose a novel Collaborative Retrieval-Augmented Generation framework, called DuetRAG, based on RAG and domain knowledge fine-tuning. This framework leverages both fine-tuned models and RAG to generate answers to questions, and a referee model is employed to determine the selection of the final answer. The aim of DuetRAG is not only to equip models with domain-specific knowledge but also to enable them to utilize external documents to obtain answers when internal knowledge is uncertain. By complementing internal and external knowledge, DuetRAG aims to enhance the robustness of the model. We provide a detailed description of the DuetRAG methodology

and its performance on various datasets in Section 3. DuetRAG outperforms existing RAG and fine-tuned models on the HotPot QA dataset (Yang et al., 2018), presenting a new pipeline for QA.

3 Related Work

Retrieval-Augmented Language Models Augmenting language models with relevant information obtained from various external knowledge bases has been shown to significantly improve the performance of various NLP tasks, including language modeling (Guu et al., 2020; Borgeaud et al., 2022; Shi et al., 2023; Lin et al., 2023) and open domain question answering (Izacard et al., 2022; Zhang et al., 2024). RAG mainly adopts the "retrieve then read" paradigm. Specifically, the input question is first used as the query, then the retrieval module retrieves relevant documents from the external knowledge base, and finally the retrieved documents and questions are merged into a complete input to generate final output. For example, RETRO (Borgeaud et al., 2022) modifies the autoregressive LM to focus on relevant documents through chunked cross-attention, thereby introducing new parameters to the model. REPLUG (Shi et al., 2023) assumes black-box access to LM and optimizes it by fine-tuning the retriever. RAFT (Zhang et al., 2024) proposes a fine-tuned data that additionally contains relevant documents and answers with reasoning chains to train language models for domain-specific open-book settings.

Finetuning for RAG Recently, related work has studied how to improve the overall performance by fine-tuning the LLM or retriever in the RAG framework. For example, RADIT (Lin et al., 2023) proposes a dual-instruction fine-tuning framework to fine-tune both the LLM and the retriever simultaneously. InstructRetro (Wang et al., 2023) pre-trains a larger autoregressive large-scale language model with retrieval function and performs instruction fine-tuning based on it. ChatQA (Liu et al., 2024) additionally proposes a context-enhanced instruction fine-tuning stage, specifically to enhance the model’s ability to perform context awareness in conversational QA. RAFT (Zhang et al., 2024) proposes a kind of fine-tuning data that additionally contains related documents and answers with reasoning chains to fine-tune LLM and improve LLM’s ability to understand the retrieved documents under the RAG framework.

Multi-Model collaboration Benefiting from the extensive knowledge exhibited by large language models, the use of multi-model collaboration to solve or improve tasks is gradually being studied. CAMEL (Li et al., 2024a) proposes a cooperative agent framework called role-playing that enables agents to cooperate on complex tasks. Du et al. (2023) and Liang et al. (2023) explore enhancing the factuality of specific tasks, e.g., translation and arithmetic reasoning, by facilitating “debates” among multiple models. Corex (Sun et al., 2023) has explored three collaboration paradigms in reasoning – debate, code review and retrieve. In addition, model collaboration can also be used between models of different sizes. Li et al. (2024b) generates prompts through a small model to guide the generation of LLM.

4 DuetRAG: Collaborative Retrieval-Augmented Generation

In this section, we introduce DuetRAG, a novel model-collaborative approach for retrieval-based argument generation, as shown in Figure 1. The DuetRAG framework relies on the synergy of three models: the Reciter, which utilizes fine-tuning to internalize a knowledge base; the Discoverer, which employs retrieval and refinement to obtain external knowledge; and the Arbiter, which evaluates and summarizes the answers from the first two models. We will first present the Problem Formalization, followed by a detailed introduction to our proposed DuetRAG.

4.1 Problem Formalization

Formally, for a specific domain task T , we have access to a training set $Train\{Q_{train}, D_{train}, A_{train}\}$. Here, Q_{train} represents the questions in the training set, D_{train} is a collection of specific documents that contain the answers to the questions in the training set, and A_{train} is the correct answers to the questions Q_{train} . During the testing phase, we have access to a test set $Test\{Q_{test}, D\}$, where Q_{test} is the set of questions from the test task, and D is the external knowledge collection. For the training set $Train\{Q_{train}, D_{train}, A_{train}\}$, D_{train} can be further classified into documents $d_1^*, d_2^*, \dots, d_i^*$ that contain direct and relevant information to the answers, and noise documents $d_1, d_2, \dots, d_j$ that are irrelevant to the answers.

4.2 Overview of DuetRAG

The core of our work, DuetRAG, involves the collaboration of three models: the question-answering model $\mathcal{M}_i$, which utilizes internal knowledge through fine-tuning based on domain expertise to answer questions; the question-answering model $\mathcal{M}_e$, which utilizes external knowledge by fine-tuning based on domain knowledge and question-answering scenarios to answer questions; and the referee model $\mathcal{M}_j$, which evaluates the answers from the aforementioned question-answering models and selects the answer with the highest confidence.

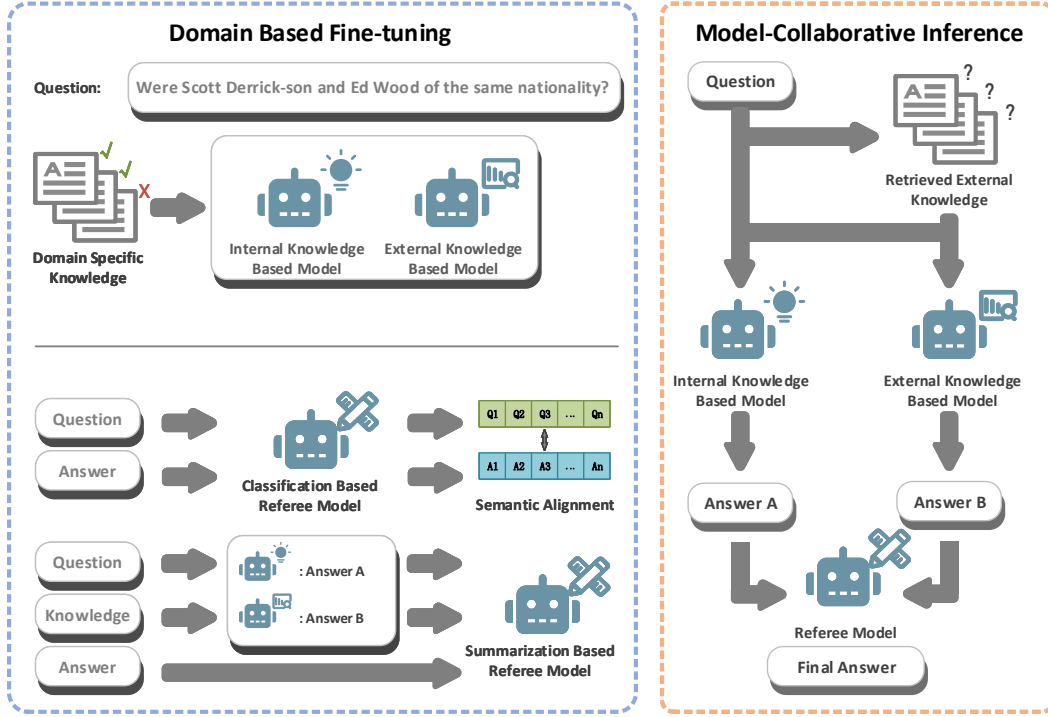


Figure 1: Overview of our DUET method.

During the Fine-tuning Stage, $\mathcal{M}_i$ learns domain-specific knowledge and recalls relevant knowledge within the model, while $\mathcal{M}_e$ learns how to retrieve and summarize answers from external documents. $\mathcal{M}_j$ is tasked with assessing the credibility of answers and generating the highest-confidence result based on existing answers.

4.3 Fine-tuning with Domain Knowledge

Our work, DuetRAG, employs a domain knowledge fine-tuning approach based on LoRA to optimize the Internal Knowledge Based Model $\mathcal{M}_i$ and the External Knowledge Based Model $\mathcal{M}_e$. For a specific domain T and training set T_{train} , we combine domain questions Q_{train} , domain knowledge D_{train} , and question answers A_{train} into a prompt q, d, a used for LoRA fine-tuning. The weights of the large model contain a vast amount of knowledge learned during the training phase, and through domain knowledge fine-tuning, the model’s knowledge relevant to the target domain will be reactivated. At the same time, the large language model possesses reasoning and summarization capabilities. Through domain knowledge fine-tuning, the model can learn to extract and answer questions related to external knowledge.

Our work on the referee model $\mathcal{M}_j$, responsible for assessing the credibility of answers and generating the final result, involves the design and implementation of two methods: classification and summarization. The classification method is inspired by the training approach of CLIP. It constructs data q, a based on the training set T_{train} to train $\mathcal{M}_j$, achieving semantic alignment between questions and answers. The summarization method utilizes the inherent reasoning and summarization capabilities of the large language model itself. By constructing prompts and fine-tuning domain knowledge, $\mathcal{M}_j$ learns to independently weigh the credibility of $\mathcal{M}_i$ and $\mathcal{M}_e$ ’s answers and respond accordingly.

4.4 Model-Collaborative Inference

During the Inference stage, DuetRAG employs a multi-model collaboration approach to obtain answers. For a given question $q \in Q_{test}$, the Internal Knowledge Based Model $\mathcal{M}_i$ attempts to provide a direct answer a_i . Simultaneously, DuetRAG utilizes a retriever to obtain relevant external knowledge documents $D_q = \{d1, \dots, dn\}$, and the External Knowledge Based Model $\mathcal{M}_e$ uses this external knowledge to answer, resulting in answer a_e . If the referee model $\mathcal{M}_j$ is implemented using the classification method, $\mathcal{M}_j$ will score the confidence levels of a_i and a_e separately based on

Table 1: Overall Performance of DuetRAG

	Accuracy(GPT-4 Eval)
LLaMA2-7B	12.7
LLaMA2-7B + RAG	9.2
DSF(LLaMA2-7B)	20.8
DSF+RAG(LLaMA2-7B)	25.4
Internal Model $\mathcal{M}_i$	23.1
External Model $\mathcal{M}_e$	32.7
DuetRAG	36.3

question q , selecting the answer a_j with the highest score. We note that the External Knowledge Based Model has access to more knowledge, so when there is only a small difference in confidence scores between the two answers, $\mathcal{M}_j$ is designed to lean towards selecting a_e . If $\mathcal{M}_j$ is implemented using the summarization method, a prompt constructed based on q , a_i , and a_e will be provided to $\mathcal{M}_j$, and the answer a_j will be obtained. This prompt contains information such as task instructions and answer preferences.

5 Evaluation

We conducted a series of experiments to validate the performance of DuetRAG. Our experiments were based on the LLAMA-7B model (Touvron et al., 2023) and the HotpotQA dataset (Yang et al., 2018). For the Ablation Study, we tested the performance of replacing modules in DuetRAG with GPT-3.5 and other methods. In future versions, we plan to evaluate DuetRAG’s performance on a wider range of task sets and assess its transferability based on different models.

5.1 Datasets

Our experiments were conducted based on the HotpotQA dataset and a collection of external knowledge from Wikipedia. In subsequent versions, we aim to include validation work on additional datasets.

HotpotQA (Yang et al., 2018) is an open-domain question-answering dataset based on Wikipedia, primarily focused on common knowledge such as movies, sports, history facts, etc. Most of the questions in HotpotQA are in the form of multi-hop questions, requiring multiple steps of reasoning and summarization to arrive at an answer. In our experiments, we used the first 5,000 data samples from the official training set containing 70,000 samples as training data, and the first 1,000 data samples from the official validation set containing 7,000 samples as validation data. The external knowledge documents were sourced from Wikipedia and the fullwiki version provided by the HotpotQA dataset.

5.2 Baselines

We consider the following baselines for our experiments. All baselines were based on the LLAMA-7B model. In future versions, we will test more base models to validate the transferability of DuetRAG.

- **LLAMA 2-7B model with 0-shot prompting:** this is the commonly used model for QA tasks. We provide clearly written instructions without any reference documents.
- **LLAMA 2-7B model with RAG(LLama2+RAG):** much similar to the previous setting, expect we include reference documents. This is a popular technique when dealing with domain specific QA tasks.
- **Domain specific fine-tuning with 0-shot prompting(DSF) :** Performing standard supervised fine-tuning without documents in context. Much similar to internal knowledge based model in DuetRAG except DSF is fine-tuned with full training data.
- **Domain specific fine-tuning with RAG(DSF+RAG):** Performing standard supervised fine-tuning without documents in context. Much similar to external knowledge based model in DuetRAG.

5.3 Results

In Table 1, we validate the performance of DuetRAG based on the aforementioned datasets and baselines. Compared to LLAMA2-7B models based on the 0-shot prompt and RAG, DuetRAG shows nearly a two-fold improvement,

Table 2: Performance of $\mathcal{M}_j$ via Multiple Approach

	Classification		Summarization	
	Alignment	Text Feature	LLaMA2-7B	ChatGPT-3.5
Acc. of $\mathcal{M}_j$	36.3	34.7	33.3	39.3
Acc. Compared with $\mathcal{M}_e$	+3.6	+2.0	+0.6	+6.6

demonstrating strong adaptability to domain-specific questions. Additionally, compared to Domain Specific Fine-tuning and RAG Based Domain Specific Fine-tuning, DuetRAG also exhibits significant improvements. Since HotpotQA is a multi-hop question-answering dataset, where retriever results may not always cover all documents containing relevant answer information, the performance of RAG-based models does not significantly surpass 0-shot prompt-based models. Furthermore, compared to Domain Specific Fine-tuned models trained on the entire training set, the performance of our $\mathcal{M}_i$ and $\mathcal{M}_e$ exceeds the former two, indicating potential overfitting of the models on excessive training data. Finally, the performance of DuetRAG surpasses that of $\mathcal{M}_i$ and $\mathcal{M}_e$, demonstrating the effectiveness of the strategy designed for $\mathcal{M}_j$.

5.4 Ablation Study

For the Ablation Study, we tested the effectiveness of employing different methods as the referee model $\mathcal{M}_j$, including classification methods based on semantic alignment, text feature-based classification methods, an ensemble method based on 0-shot LLaMA-7B, and an ensemble method utilizing ChatGPT-3.5 (in Table 2). We can see that the classification method based on semantic alignment improved accuracy by 3.6 percent, surpassing the direct text feature-based classification method. Although the Summarization Method using ChatGPT-3.5 demonstrated significant performance enhancements, these improvements are primarily attributed to the powerful summarization capabilities of ChatGPT due to its large-scale training. However, similar summarization methods applied to the 0-shot LLaMA2-7B model did not yield particularly favorable results, likely due to the limited summarization capabilities of LLaMA2-7B itself.

6 Conclusion

DUET is a novel multi-model collaboration framework designed to enhance the performance of question-answering models in specific domains with few training samples and complex environments. We have identified several key design decisions, such as simultaneously using an Internal Knowledge-Based Model and an External Knowledge-Based Model to generate answers, and using a third model to generate the answer with the highest confidence level. Our evaluation on HotpotQA highlights the significant potential of DUET. Looking ahead, we anticipate that Retrieval-Augmented Generation (RAG) within the domain will continue to attract attention in both industry and academia. Unlike general RAG, our work addresses the practical scenario where LLMs are tasked with answering questions using domain-specific knowledge. Meanwhile, we will continue to update this work and validate the reliability and transferability of DUET on more application scenarios and base models.

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